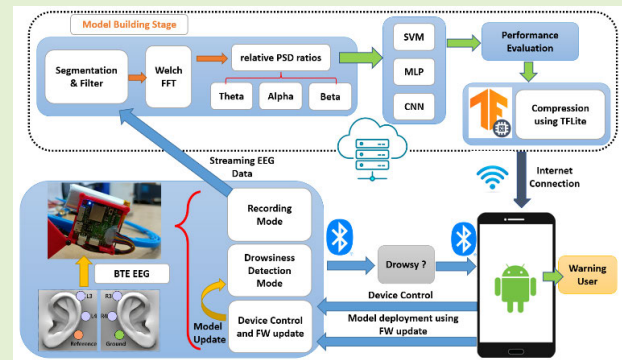


Behind-the-Ear EEG-Based Wearable Driver Drowsiness Detection System Using Embedded Tiny Neural Networks

Ha-Trung Nguyen, Ngoc-Dau Mai, Boon Giin Lee^{1b}, *Senior Member, IEEE*,
and Wan-Young Chung^{1b}, *Senior Member, IEEE*

Abstract—Driver drowsiness is a major factor that leads to traffic accidents and loss of life. A popular drowsiness detection (DD) method is based on machine learning (ML) and monitoring electroencephalogram (EEG) signals from the scalp. However, the setup is usually bulky and too obtrusive for use in daily life. In addition, the data processing unit is usually placed in large hardware such as cloud servers and personal computers that receive the EEG signal from wearable sensors wirelessly. This results in a cumbersome system that lacks mobility and suffers high latency and power consumption because of wireless connection. The behind-the-ear (BTE) EEG is a novel approach that promises a more convenient and practical solution than scalp-EEG. Recent advances in tiny ML (TinyML) mean that on-device ML-based driver DD (DDD) could also be realized to avoid the limitations mentioned above. This article presents the design of a novel BTE EEG-based DDD system that uses a wearable headband device and runs built-in neural network (NN) models. The device collects EEG signals from four proposed BTE locations that are then preprocessed. Welch's method is applied to extract the relative power spectral density (rPSD) ratio of three EEG bands (theta, alpha, and beta). Two NN models [multilayer perceptron (MLP) and convolutional NN (CNN)] were trained and compared to a support vector machine (SVM) before deployment in the embedded device. Then, a benchmark test was performed to evaluate the on-device performances of these models. The test results established the practicality and feasibility of the proposed system for on-device DDD.

Index Terms—Drowsiness detection (DD), edge computing, electroencephalogram (EEG), microcontroller, tiny neural networks (NNs), wearable system.



I. INTRODUCTION

DROWSINESS is a transitional state between alertness and sleep with reduced cognitive and processing abilities. Drowsiness can significantly affect the performance of a driver

and lead to traffic accidents. Around the world, thousands of deaths and injuries have been reported each year due to driver drowsiness. The National Highway Traffic Safety Association (NHTSA) reported that driver drowsiness was the cause of 2.3%–2.5% of all fatal crashes in USA from 2011 to 2015 [1]. Thus, driver drowsiness has become a critical concern for traffic safety, and the development of driver drowsiness detection (DDD) systems that can detect drowsiness and improve the driver's alertness has received attention as a possible solution. Previous approaches to developing DDD systems can be categorized according to the measurement method, such as vehicle-based measures (e.g., deviation in lane position and steering wheel movement) [2], [3], behavioral measures (e.g., yawning, percentage of eyelid closure over the pupil over time (PERCLOS), and head pose) [4], [5], [6], and physiological signal measures [e.g., electrocardiogram (ECG), electromyography (EMG), electrooculography (EOG), and electroencephalogram (EEG)] [7], [8], [9], [10]. EEG-based DDD is one of the most reliable and effective methods, and it has gained recent attention because of the direct correlation between the EEG signal and brain states [11], [12]. Various

Manuscript received 28 June 2023; revised 4 August 2023; accepted 15 August 2023. Date of publication 31 August 2023; date of current version 2 October 2023. This work was supported by the National Research Foundation of Korea (NRF) Grant funded by the Korea Government through Ministry of Science and ICT (MSIT) under Grant NRF-2019R1A2C1089139. The associate editor coordinating the review of this article and approving it for publication was Dr. Lin Wang. (Corresponding author: Wan-Young Chung.)

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by Pukyong National University Institutional Review Board under Application No. 1041386-202003-HR-11-02.

Ha-Trung Nguyen, Ngoc-Dau Mai, and Wan-Young Chung are with the Department of Artificial Convergence, Pukyong National University, Busan 48513, South Korea (e-mail: nguyenhattrung411@gmail.com; ngocdaumai95@pukyong.ac.kr; wychung@pknu.ac.kr).

Boon Giin Lee is with the Nottingham Ningbo China Beacons of Excellence Research and Innovation Institute, School of Computer Science, University of Nottingham Ningbo China, Ningbo 315100, China (e-mail: boon-giin.lee@nottingham.edu.cn).

Digital Object Identifier 10.1109/JSEN.2023.3307766

EEG-based portable DDD systems have been proposed for real-world application, as reviewed by LaRocco et al. [13]. Their review showed that a change in EEG band power is a commonly used indicator for DDD. In particular, the onset of drowsiness tends to be accompanied by a decrease in the power of alpha and beta bands and an increase in theta band [14], [15].

Many previous studies on EEG-based DDD methods measured EEG signals from multiple locations on the scalp using the 10–20 electrode system [10], [16], [17], [18]. A critical influencing factor for the EEG data quality is the impedance, which captures how poorly signals flow to the electrode. Because most of the scalp is covered by hair, this can significantly increase the impedance, so the collected signal can have more electrical noise than the actual signal. Therefore, scalp-EEG systems need to utilize a wet or active electrode to guarantee the signal quality, which also makes the system bulky, uncomfortable, and complicated for real-life applications. Another solution for hair-covered scalp problem is using a brush-type electrode, which has small bristles that can easily penetrate hair and create a larger contact area with the scalp, leading to enhanced signal quality. However, these brush-type electrodes can cause skin irritation or discomfort for some individuals. To make scalp-EEG-based DDD systems more portable and unobtrusive, certain studies have employed fewer electrodes, mostly placed in the occipital region (at the back of the head) [23], [49], [50]. This region was usually chosen because of its prominent display of alpha activity and its involvement in visual processing, which is often linked to drowsiness. Nevertheless, when used for detecting drowsiness of driver, using measurement points in this region can cause discomfort while relying on the seat while driving. The pressure exerted pushes the electrodes forcefully into the head, causing discomfort or, in the case of using brush-type electrode, even resulting in scalp abrasions. In addition, the force applied while leaning against the seat can also damage electrodes.

A potential alternative for overcoming the limitations of scalp-EEG systems is to measure the EEG signal around the ears [19], [20]. Ear EEG electrodes are only placed in or near the outer ear, which allows for much greater visibility and mobility of the wearer and better skin-to-electrode contact for improved signal quality. However, this approach may result in a reduced signal amplitude and fewer brain regions measured [20], [21], [22]. Ear EEG systems can be categorized into two groups based on the measurement location: 1) in-ear (i.e., electrodes inside the ear canal) and 2) behind-the-ear (BTE) (i.e., electrodes close to the ear and usually behind the earlobe). Guermandi et al. [48] presented a wireless system in the form of an earbud for EEG acquisition and processing, demonstrating its ability to detect brain responses using in-ear EEG. The device's printed circuit board (PCB) size is $15 \times 16 \text{ mm}^2$ and it can collect data from one in-ear EEG location. The system has the capability to stream raw EEG data or perform onboard processing to transmit only the results. Related to the EEG-based DDD application discussed in this work, previous studies, such as [43] and [44], have

also explored in-ear EEG usage. Although both studies demonstrate promising results in using in-ear EEG to detect drowsiness (88.8% and 88.9% of accuracy), the systems they used for research were not portable and unobtrusive for real-life application. Pham et al. [52] introduced a portable system that utilizes BTE EEG to detect microsleep events, which often occurs during an extremely drowsy state. This system can collect and stream data from eight measuring locations and stream them to PC for processing. The researchers also highlighted the relationship between ear EEG measurement points and the orexin system, which functions as a network promoting wakefulness throughout the central nervous system by stimulating neuron activity in the midbrain, cerebrum, and visual cortex. These neural activities are represented by brain waves, such as rapid beta (b) and alpha (a) waves when the brain is alert and conscious, and slower theta (u) waves when drowsiness or sleepiness is experienced. Overall, although in-ear EEG offers a more unobtrusive alternative to BTE EEG, it typically provides fewer measuring locations (usually 1–2 locations in most studies), which can potentially impact the system's robustness and performance. In addition, there are concerns about the use of in-ear EEG designs that typically require sealing the ear canal to allow the electrodes to contact measuring locations, as seen in [43], [44], and [48]. This can significantly reduce the driver's awareness of the surrounding environment, including important auditory cues such as honking horns, emergency vehicle sirens, or other vehicles' engine noises. On the other hand, the use of BTE EEG can avoid these issues. It is also worth noting that research on the development of ear EEG-based DDD systems is still limited, and most studies have focused on in-ear EEG signals.

Most conventional low-power wearable EEG-based systems consist of two main parts: a wearable device for acquiring the EEG signal and computer-based hardware such as a PC, smartphone, or smartwatch for data processing [10], [16], [17], [23]. The first part comprises an EEG signal acquisition module, a wireless communication module [e.g., Wi-Fi or Bluetooth/Bluetooth low energy (BLE)], and low-power microcontroller unit (MCU) containing one or more processor cores, limited memory, and programmable input–output peripherals. This reduces the hardware costs and minimizes the power consumption and size for improved wearability. This part is responsible for acquiring the raw EEG signal and transmitting it wirelessly to the computer-based hardware, which handles the remaining stages such as preprocessing, feature extraction (FE), and drowsiness detection (DD). For wireless systems, increasing the length of the transmitted packet can increase the power consumption and propagation delay, which can significantly increase the total delay of the system. The length of the transmitted packet is mainly affected by the number of used EEG channels and the byte resolution for each EEG channel. To address the above limitations, one approach is to move all the computational stages to the wearable EEG device (on-device process). This solution makes the computation independent of the wireless communication latency, so the wearable device only needs to transmit the classification results

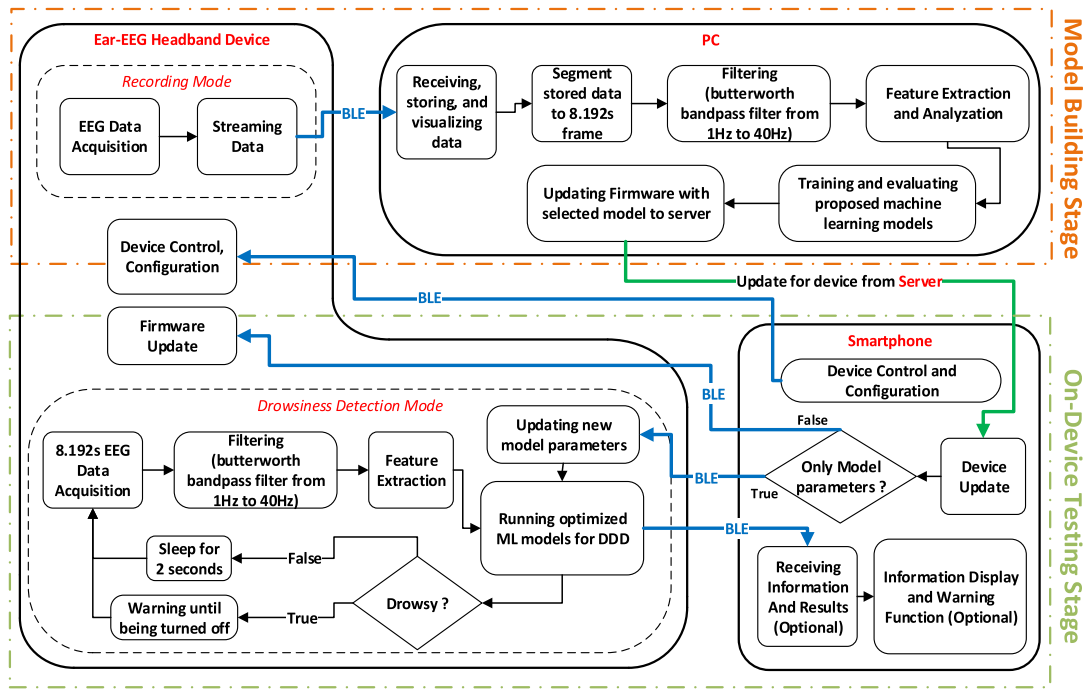


Fig. 1. System design overview with two stages: MB stage and ODT stage.

(1 or 2 bytes) rather than the raw EEG signal or it may not even need to send anything in case the user interface is on the wearable device. This approach also reduces or even eliminates the power consumption of wireless communication. However, MCUs are memory-constrained systems with a limited processing frequency of less than 1 GHz, which has inhibited the applicability of this approach. Consequently, it can make on-device processing idea more challenging when implementing complicated detection algorithms, especially deep-learning approaches, which recently gather more attention from researchers as its potential in achieving state-of-the-art performance in various tasks.

Tiny machine learning (TinyML) [24], which is dedicated to offering solutions to optimizing models to run on MCU platforms, has risen as a promising approach to fill the gap between microcontrollers and machine learning (ML), especially deep learning. Depending on the system architecture design, TinyML can provide several key benefits.

- 1) *Privacy*: In some cases, processing data at the edge device instead of transmitting to a central device (e.g., cloud) may prevent possible violations of privacy or vulnerability to interception.
- 2) *Latency*: Because the entire processing phase runs on the edge device, wireless latency is not a concern. Thus, decision-making tasks can be processed in real time at the edge device.
- 3) *Energy efficiency*: Even though the processing stage is energy-intensive, the power consumption of data transmission can be significantly higher.

We present a design of a wearable DDD system that utilizes the BTE EEG signal and the TinyML concept for on-chip ML. In particular, the main contributions of this article are given as follows.

- 1) Propose a novel design of an on-device BTE-EEG-based DDD headband system.
- 2) Investigate and analyze the feasibility of utilizing BTE EEG signals with the feature of relative power spectral density (rPSD) ratios from three specific bands (theta, alpha, and beta) for DDD.
- 3) Propose two tiny NN models to implement on a resource-constrained MCU-based device for DD using extracted features of BTE EEG.
- 4) Validate the performance and effectiveness of two proposed models for on-device DDD.

The rest of this article is organized as follows. Section II presents the proposed system. Section III presents the methodology of experiments performed to evaluate the feasibility and practicality of the proposed system as well as the performance of different deep-learning and ML models. Section IV presents the experimental results of the models and their deployment in on-device tests. Section V concludes this article.

II. PROPOSED SYSTEM

A. System Overview

The proposed BTE EEG-based on-chip DDD system consists of two devices: a BTE EEG headband device and an Android smartphone. The system design overview is shown in Fig. 1. The design process had two stages: model building (MB) and on-device testing (ODT). In the MB stage, the headband device was operated in the recording mode. It acquired and streamed data to a computer for storage and processing. Then, the computer preprocessed the stored data and extracted features from the EEG bands for classification. To perform the FE task, two deep-learning models were trained for comparison with a conventional ML model. The ODT stage involved evaluating the signal processing, FE, and models

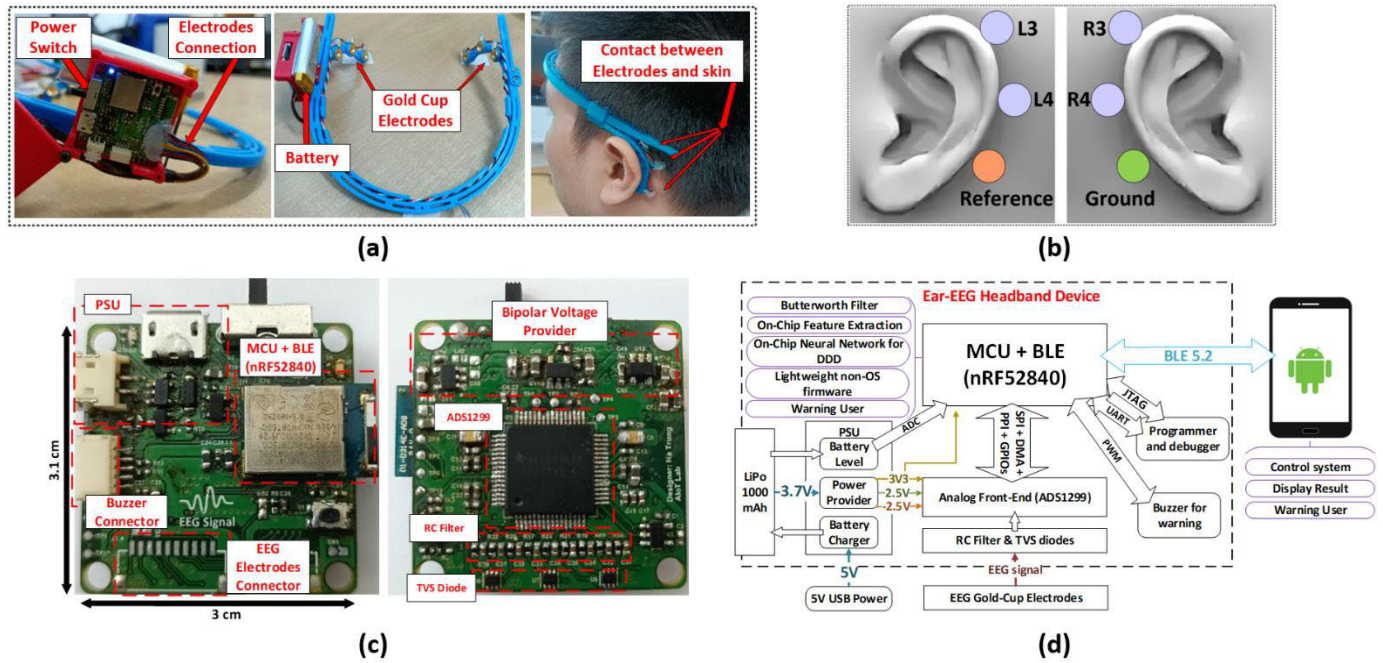


Fig. 2. (a) Overview of the proposed headband device. (b) EEG measurement locations behind the ear (left and right ears). (c) Back and front sides of PCB part of the proposed headband device. (d) Block diagram of the proposed device.

for DDD performance. An Android smartphone application was developed with features that help users interact with the headband device, such as device control and information display. For convenience, the smartphone could deploy pre-trained model parameters to the headband device or update its firmware from the server. When drowsiness was detected, the system had the option of warning the user through the headband device or smartphone.

B. BTE EEG Headband Device and Electrode Locations

The frame of the headband device was 3-D-printed from a flexible material to fit a human head comfortably and to be adjustable. A PCB is attached to the right side of the headband with a battery (3.7 V, 300 mAh). Fig. 2 shows the design of the headband device, which combines a low-power, high-performance, and low-cost MCU for edge device ML with a high-precision analog front end (AFE) for EEG data acquisition. The PCB has dimensions of $3.1 \times 3 \times 0.5$ cm and contains three blocks: the sensor collecting unit (SCU), processor and wireless transceiver unit (PWTU), and power supply unit (PSU).

The SCU is responsible for EEG signal acquisition and uses the very accurate AFE integrated chip (IC) ADS1299, which features eight low-noise and eight 24-bit resolution simultaneous-sampling analog-to-digital converters (ADCs). This chip is designed for EEG applications by Texas Instrument [25]. It uses the serial peripheral interface (SPI) method as communication for signal data retrieval. ADS1299 also has a data-ready pin to inform the SPI master when the EEG signal data are converted to enhance the data reading efficiency. The ADS1299 not only provides a wide range of data rates, ranging from 250 SPS to 16 kSPS for each channel, at low power but also offers a programmable gain amplifier (PGA)

with a maximum gain of 24. It can provide better signal quality with less noise compared to its predecessor, ADS1298, which has been widely used in many EEG studies. However, the tradeoff is that it consumes more energy. The analog supply of ADS1299 is connected to a bipolar source ranging from -2.5 to 2.5 V supplied by the PSU. Resistor–capacitor (RC) low-pass filters with a cutoff frequency of 72.34 Hz are installed in the signal path of each EEG electrode input for antialiasing. A transient voltage suppression diode array is utilized to prevent electrostatic discharge (ESD) noise upon contact with human skin.

The PWTU uses the multiprotocol Bluetooth 5.3 system-on-chip nRF52840 from Nordic Semiconductor. It has a 32-bit ARM Cortex-M4F CPU core with a maximum clock frequency of 64 MHz and extensive memory with 256-kB RAM/1-MB flash memory. In addition, nRF52840 offers a rich and varied set of peripherals, including an SPI port with a maximum speed of 4 MHz to communicate with ADS1299 and a programmable peripheral interface (PPI) that we used to operate the data-ready pin of ADS1299. Furthermore, nRF52840 meets BLE 5 specifications and can achieve up to four times the range and twice the on-air data rate (up to 2 MBs from 1 MBs) compared with BLE 4.

The PSU uses the battery charge management controller MCP73831 for high efficiency. The low-dropout regulator AP2112 is utilized to convert the input voltage to the operating voltages of the SCU and PWTU. The bipolar voltage provider subsystem for ADS1299 comprises three components: TPS60403 for inverting the input voltage, TPS72325 for converting the inverted input voltage to -2.5 V, and TPS73225 for converting the input voltage to 2.5 V.

The electrodes used in this study are passive dry-contact gold cups, ensuring a consistent and reliable signal. These

electrodes are attached to the headband frame at the desired locations for data collection. The headband frame helps maintain good contact between the electrodes and the skin, as shown in Fig. 2(a). Based on the electrode positions of the cEEGrid, which is a flex-printed and multichannel sensor array placed around the ear using adhesive for researching sleeping stages and epilepsy [26], we used positions R3, R4, L3, and L4 for measuring EEG signals, as shown in Fig. 2(b). The ground electrode is placed on the right mastoid, and the reference electrode is placed on the left mastoid.

C. Firmware Design and BLE Settings

Our proposed system has two modes: recording (REC) and DD. In the REC mode, EEG data are collected and streamed to the computer. Because the AFE chip ADS1299 samples EEG data at a rate of 250 SPS up to 16 kSPS with eight 24-bit channels, it can produce 48 kbits to 384 000 kbits/s. The BLE stacks of nRF52840 can transmit four to six data packets per connection event, and the delay between two consecutive packets (i.e., interframe space) is about 150 μ s. Each connection interval (CI) can last a minimum of 7.5 ms, with an increase in the interval coming with the tradeoff of increased power consumption. In a real-world environment, the latency of BLE is affected by various factors such as the transmitting distance and transmitting errors causing the transmitter to resend the packet. Thus, the received data sampling rate can be reduced if the wireless gathering firmware is designed linearly in a single loop (i.e., collect data, transmit data, and then repeat). Therefore, to secure the EEG data sampling rate after transmission to the computer, the firmware needs to send data through BLE and acquire EEG data independently. In the REC mode, we use the PPI to retrieve data from ADS1299 by SPI communication, which is triggered directly by the data-ready pin. At the same time, the main loop program of nRF52840 handles data transmission. The PPI enables peripherals to interact autonomously independent of the CPU to decrease the computational load on nRF52840. A ring buffer is utilized to transfer the data obtained from the ADS1299 data acquisition task to the BLE transmission task. The ring buffer ensures that the sampled data are not lost even when the BLE task is occupied or busy. We also take advantage of the direct memory access (DMA) features of nRF52840 to enhance the SPI communication with ADS1299. Fig. 3 shows the design of the firmware in REC mode. The DD mode runs tasks for DD with the deployed model and treatment sequentially, as shown in Fig. 1. A 2-s sleep state is added to the running sequence when drowsiness is not detected to reduce power consumption.

The BLE settings for both modes utilize the notification mechanism for BLE transmission. In the REC mode, the data transmission needs to be as fast as possible, so we utilize a minimum CI of 7.5 ms. However, it is important to note that this high-speed transmission also consumes the highest power for BLE. On the other hand, in the DD mode, the device takes some time to process on-chip DD before sending the result. To optimize power consumption, the CI is set to 100 ms in this mode. The PHY link of BLE is set to 1 Mb/s type.

The BLE packet size for REC mode is 14 or 26 bytes (four channels or eight channels), including two first bytes for packet

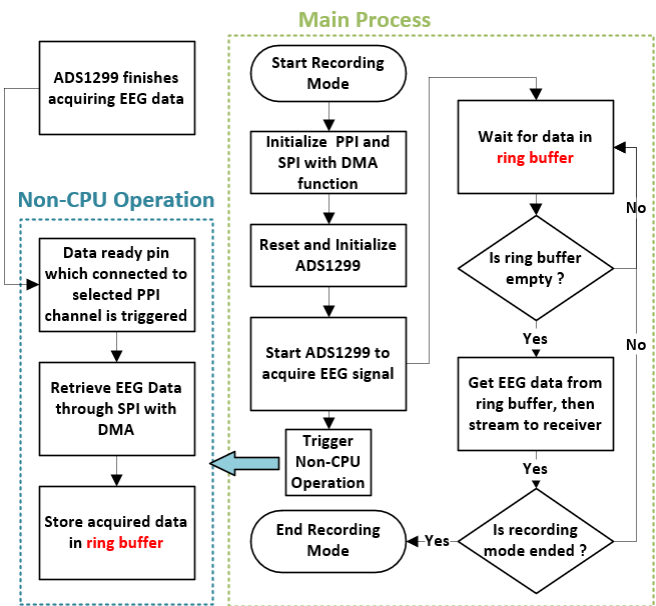


Fig. 3. Function block diagram of firmware in the recording mode.

counter and the remaining for EEG data. For the DD mode, the packet size is 4 bytes, including two first bytes for packet counter, 1 byte for prediction results, and 1 byte for accuracy percent.

D. Smartphone Application and Server for Model Deployment

To provide a better user experience, we have developed a smartphone application that enables users to control the headband device. With this app, users can easily switch between two modes: recording mode or DD mode. In the DD mode, users can choose between closed- or open-loop mechanisms. In the closed-loop mechanism, the device operates without sending any results to other devices and uses an onboard buzzer to warn the user when detecting drowsiness. This helps conserve energy by eliminating almost all BLE communication. In the open-loop mechanism, the device sends results and information to other devices, such as a smartphone or car system, via BLE communication. The driver warning module is then executed by the other device and various components in the car can work together to enhance user alertness. For example, the car system can automatically control the air conditioning to lower the temperature slightly, which can help the driver feel more alert.

In real-world applications of ML, models are continually developed and trained to enhance their performance. As a result, it is important to have a mechanism for updating devices with newer, higher performing models. Our system includes firmware updates that can be sent to the headband device via the smartphone application, ensuring that users always have the latest models. To update the firmware, the application downloads the up-to-date model from a server using any available internet connection, such as Wi-Fi or 4G LTE. Next, the application initiates a firmware update through BLE communication to the headband device.

III. MATERIAL AND METHODOLOGY

A. Experimental Protocol and Data Annotation

For the experiments, we recruited 12 subjects ranging in age from 24 to 38 years old. They were physically and mentally healthy without any optical or sleep problems, and they possessed valid driving licenses. The experiments had two measurement sessions. The first session measured the EEG signal of subjects when they were driving in their most alert state. The second session measured the EEG signal of the subjects when they were easily likely to become drowsy while driving. The NHTSA reported that traffic accidents due to driver drowsiness occurred most frequently between 12 A.M. and 6 A.M. or in the late afternoon [29]. In addition, the most wakeful period for the majority of people who have a normal circadian cycle is between 9 A.M. and 11 A.M. [30]. Therefore, the first session was conducted in the morning at 10 A.M. Then, subjects were required to wait until midnight (12 A.M.) without sleeping for the second session. Each session lasted about 1 h.

In each session, subjects performed driving tasks on the commercial driving simulator software BeamNG.drive by controlling a set of a Logitech steering wheel and acceleration and brake pedals. The selected map included highway roads in a rural area with a low traffic volume. During the monotonous driving experiment, subjects were subjected to various natural cognitive loads while driving, such as controlling the vehicle to avoid collision with other vehicles and deceleration to stop the vehicle at a red light. Subjects were told to maintain a vehicle speed up to about 50 mph. Two cameras were set up: one to record the subjects' behavior and the other to record the subjects' facial expression and eyelid states. Fig. 4(a) shows the experimental setup.

Before the experiments, subjects had a few minutes to familiarize themselves with the simulator. Then, the Berger effect, which is mentioned in Section IV-A, was tested to ensure the EEG signal quality. Each subject was instructed to avoid alcoholic, caffeinated, and energy drinks and smoking for one day before the experiments. The consumption of drowsiness-triggering medication was also prohibited.

To confirm the detected driving state, subjects were asked to rate their level of drowsiness/alertness at the beginning and end of each session according to the Karolinska sleepiness scale (KSS) [31]. The KSS is a widely used subjective evaluation method of sleepiness and comprises a nine-point scale ranging from 1 (full alertness) to 9 (extreme drowsy). The reliability and validity of the KSS have been proven in previous EEG studies [32], [33]. The morning sessions showed that all subjects rated themselves as 2–4 on the KSS at the beginning and 3–5 at the end, which indicates a normal driving state. In the midnight sessions, subjects rated themselves as 6–7 at the beginning and 8–9 at the end, which indicates drowsiness. The Wierwille scale (WS) [34] was also applied to measure drowsiness/sleepiness using the video recorded from the two cameras. The WS is widely used to obtain the ground truth for video-based drowsiness evaluation [16], [23]. First, the video recorded by the cameras was divided into 30-s clips. Three observers then evaluated the clips independently and labeled them as normal or drowsy according to the WS conditions

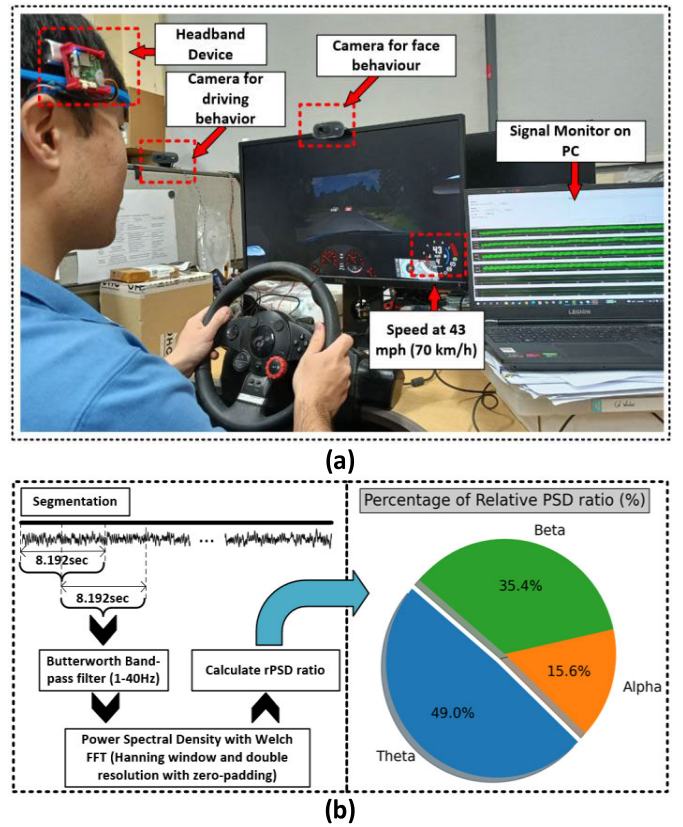


Fig. 4. (a) Example of experiment setup for gathering data. (b) SP&FE on an 8.192-s EEG data epoch.

TABLE I
WIERWILLE-BASED SCALE FOR LABELING DATA

State	Driving Behaviors
Normal	• Normal fast eye blinks, often reasonably regular. • Apparent focus on driving with occasional fast sideways glances. • Normal facial tone. • Occasionally head, arm, and body movements.
Drowsy	• Increase in duration of eye blinks. • Possible increase in the rate of eye blinks. • Rubbing the face or eyes, scratching, moving restlessly in the seats. • Slower eyelid closure, the facial tone may decrease, staring at a fixed position, yawning.

given in Table I. The final decision was based on majority voting. The KSS and WS methods were used as the ground truth for the labeled data.

B. Signal Preprocessing and Feature Extraction

For the MB stage, the acquired BTE EEG signals were processed on a computer. We segmented the raw EEG data of each channel of each subject into 8.192-s epochs (2048 data points) with an overlap rate of 50% to build the epoch dataset for each driving state, which comprised a [channel $\times$ time $\times$ epoch] matrix. Then, a fourth-order Butterworth bandpass filter at 1–40 Hz was implemented on each data epoch to eliminate unwanted noise, particularly power-line noise (60 Hz). For FE, we used the features related to the PSD of three EEG bands:

theta (4–8 Hz), alpha (8–13 Hz), and beta (13–30 Hz). Instead of the absolute PSD values, the rPSD was utilized to reduce individual differences and enhance the generality of the input feature data for the classification models. This kind of feature was used widely in previous research for scalp-EEG. For each data epoch, Welch's method was applied, which uses a moving window technique where the fast Fourier transform (FFT) is computed in each window and the average is taken over all windows to obtain the PSD values. We used the Hanning window function, which has a good frequency resolution for FFT and reduces spectral leakage. The window size was set to 512, and the overlap rate was 50%. To double the frequency resolution, the zero padding of 512 points was applied to each window frame for FFT. The rPSD ratio of each desired EEG band was calculated by dividing the sum of the PSD values in frequency bins covering the range of a given EEG band by the sum of PSD values in frequency bins covering the range of all three EEG bands

$$\text{rPSD}_i = \frac{\text{PSD}_i}{\sum_{j=1}^3 \text{PSD}_j} \quad (1)$$

where rPSD_i and PSD_i are the rPSD ratio and total PSD value of the i th band, respectively ($i = 1, 2$, and 3 corresponding to theta, alpha, and beta, respectively).

Finally, we obtained a set of three extracted rPSD ratio features for an 8.192-s epoch. This procedure is shown in Fig. 4(b). After the signal preprocessing and FE (SP&FE) task was performed for all remaining epochs, we obtained the extracted feature dataset comprising a [channel $\times$ feature $\times$ epoch] matrix for each subject. For the ODT stage, all steps except segmentation were embedded in the microcontroller because the headband device performed the signal processing and detected drowsiness immediately after collecting an 8.192-s data epoch, as shown in Fig. 1. To ensure the uniformity of the SP&FE task on both the PC and MCU in the two stages, the fourth-order Butterworth filter and FFT-based FE using Welch's method were developed in C/C++ and checked by comparing the output with the widely used Python library *SciPy* (*scipy.signal.butter* and *scipy.signal.welch*, respectively).

C. Classification

1) *Neural Network-Based Models*: In this study, we considered two deep-learning models for evaluating the DDD performance of our proposed system. The multilayer perceptron (MLP) is a feedforward neural network (NN) that learns via the backpropagation method [35]. MLPs are fully connected, meaning that each neuron in one layer is connected to all neurons in the next layer. It comprises three main parts: a layer of neurons that serves as the input, one or more hidden layers that receive a set of weighted inputs and then generate the output through activation functions, and the output layer that predicts the outcome. Convolutional NN (CNN) is a type of NN frequently used to categorize multidimensional data, including photos, videos, and multichannel biosignals. CNN is advantageous because it requires less manual labor or preprocessing than other deep-learning models. The key component of a CNN is the convolutional layer, which applies filters to the input data and captures local features. While MLP

and CNN have different architectures, they can be combined in hybrid models to take advantage of the strengths of both networks. For instance, a CNN can be followed by a fully connected MLP for classification or regression or an MLP can be used for FE before passing the output to a CNN for further processing.

Our proposed MLP consists of two hidden layers with the first layer containing 16 nodes and the second layer containing two nodes. Each hidden layer is followed by a batch normalization to improve the performance and stability during training, and an activation rectified linear unit (ReLU) activation function to ensure nonlinearity. The input data were passed through a flatten layer before reaching the first hidden layer.

For the CNN-type model, we proposed a compact architecture. In the first convolutional layer, we intended to make the model learn the feature for each pair of extracted subband's rPSDs in each channel of the input data (four channels $\times$ three rPSDs). Therefore, first, we duplicated the rPSD of alpha band (F1) and put it to the last in each channel input data (see Fig. 5). Then, the new input dataset with the size of (four channels $\times$ four rPSDs) was passed through a 2-D convolutional layer deployed with a kernel size of (1, 2), a stride of 1, and eight filters. In the next layer, the 2-D depth-wise convolution layer (kernel size of (4, 1) and depth parameter $D = 1$) was utilized to learn spatial filters to combine the information of all the channels of each feature map. We also apply batch normalization along the feature map dimension before applying the ReLU nonlinearity.

In the final layers of both proposed CNN and MLP models, the *softmax* activation function was used as the output layer for classification. *Softmax* is a logistic function widely used for binary classification, where the model outcome is interpreted as the probability of data belonging to different classes with a sum equal to 1. Dropout layers, which randomly set input units for the following layers to zero with a frequency rate of 0.5 at each step during training time, were also applied to both MLP and CNN to prevent overfitting. Note that the dropout layer was only applied during the model training process. The detailed architectures of MLP and CNN are shown in Fig. 5. Both the proposed models were trained utilizing a categorical cross-entropy loss function

$$L(\hat{y}, y) = - \sum_{i=1}^K y_i \log(\hat{y}_i) \quad (2)$$

where y are true labels, K denotes the number of classes, and $\hat{y}$ is the prediction of the model.

2) *Support Vector Machine*: To serve as a baseline, the support vector machine (SVM) was also trained for comparison with the two deep-learning models. SVM is a widely used ML model in EEG-based DD studies [16]. A conventional binary SVM classifier finds a hyperplane that separates two classes and has the most significant margin for each class [36]. By using kernel function methods and the regulation parameter C , the SVM can be used to deal with nonlinear datasets by mapping them into a higher dimensional feature space [36]. In this study, the radial basis function was utilized as the

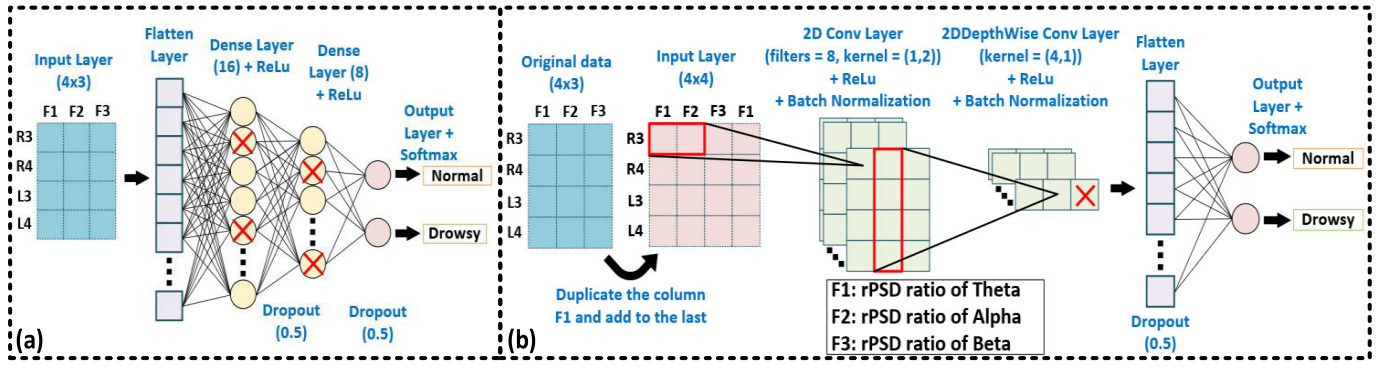


Fig. 5. Proposed (a) MLP model and (b) CNN model for on-device DD.

nonlinear kernel for SVM with $C = 1000$. This is a popular kernel function for kernelized learning algorithms.

D. Deployment on the MCU

In the ODT stage, the first step was to implement the SP&FE task on the MCU. The output on the device was checked to guarantee its similarity with that on the computer. To deploy our pretrained models on the headband device, we utilized the TensorFlow Lite for Microcontrollers (TFLM) framework. TFLM is designed to run TensorFlow-based NN models on MCUs and other devices with only a few kilobytes [37]. TensorFlow Lite converts a trained TensorFlow-based model into a smaller and optimized ML model format called FlatBuffer (FB) [38]. Then, this format, which includes 32-bit floating-point weights, can be converted into C/C++ FB code for deployment on embedded devices. Alternatively, before conversion to C/C++ code, the quantization technique can be applied to reduce the model size and improve hardware latency with little degradation of model accuracy. This technique uses the min-max algorithm to convert weights from 32-bit floating-point to 8-bit integer precision. This algorithm uses the minimum and maximum values of weights and activation tensors. It uses integer arithmetic to allow for more effective implementation than floating-point inference, and it is based on a representative convention used by Google for quantized models [39]. The quantization process maps integers q to real number r by using the two constant parameters S (scale) and Z (zero point)

$$q = \frac{r}{S} + Z. \quad (3)$$

S is an arbitrary positive real number, and it affects how precise the quantized value q is linearly distributed around the “zero” point Z . Then, the C/C++ FB is stored in the target MCU memory to run along with the TFLM library. When the target MCU begins making inferences, the TFLM library interprets the C/C++ FB data to extract information about weights and operations used in the optimized models. Because of the FB technique, the model on the target MCU can be quickly upgraded and updated by replacing the FB-converted model weights. Therefore, instead of a full firmware update of the proposed system, only the new model weights need to be updated. These are stored in flash memory and are transmitted

TABLE II
COMPARATIVE ANALYSIS OF INPUT-REFERRED NOISE ACROSS AVAILABLE SAMPLING RATES OF THE PROPOSED DEVICE

Sampling Rate	INPUT-REFERRED NOISE (μV_{rms}) FOR EACH PGA GAIN						
	Gain = 1	Gain = 2	Gain = 4	Gain = 6	Gain = 8	Gain = 12	Gain = 24
250 SPS	1.1	0.54	0.33	0.21	0.18	0.16	0.11
500 SPS	1.51	0.78	0.49	0.31	0.27	0.25	0.18
1 kSPS	1.92	1.16	0.59	0.43	0.4	0.31	0.24

from the Android application to the headband device through BLE after the Android app receives them from the server. The model deployment process is shown in Fig. 6.

IV. EXPERIMENTAL RESULTS

A. Electrical Characterization of the Proposed Device

A comprehensive evaluation was conducted to assess the wireless sampling capabilities of the device in the REC mode. It is noted that the evaluation is conducted with fully eight channels. The evaluation involved systematic testing of various sampling rates, starting from 250 SPS and gradually increasing. Each sampled eight-channel data point was packaged with a 2-byte counter variable and seamlessly integrated into the ring buffer for delivering to the BLE task. By carefully examining the sequential order of these counter variables, any occurrences of data loss were effectively detected. Furthermore, continuous monitoring of the available space in the ring buffer was carried out during operation. If the ring buffer consistently reached its maximum capacity, it indicated that the processing speed of the BLE task was insufficient to keep up with the data collection rate, resulting in data loss. After testing, we observed that the proposed device in the REC mode can do streaming data stably with a sampling rate of AFE from 250 SPS to 1 kSPS without any data loss.

Then, we keep considering testing the device in REC mode with single-ended montage (all negative input connected together to a reference electrode) with three configurations of sampling rates: 250 SPS, 500 SPS, and 1 kSPS, which provides a -3 -dB cutoff frequency of 65, 131, and 262 Hz, respectively. It is worth mentioning that by adjusting the data rate and PGA setting of the ADS1299, the noise performance

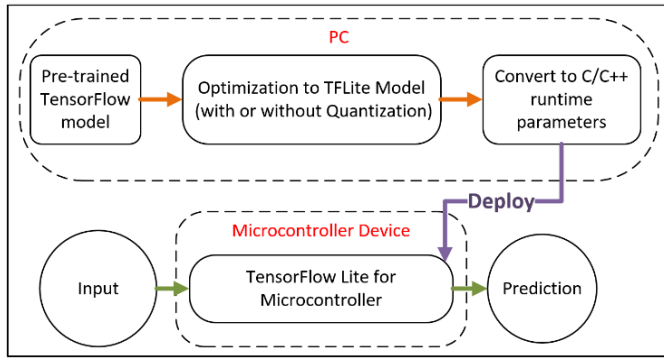


Fig. 6. On-device NN model deployment process description.

can be optimized [25]. Lowering the data rate increases the level of averaging, leading to a corresponding reduction in noise. Conversely, increasing the PGA value helps in reducing the input-referred noise. To examine these configurations, the system noise is measured by shorting the inputs of the AFE and integrating over the standard EEG band (0.5–100 Hz), as shown in Table II. For the sampling rate of 250 SPS, the noise is measured at the maximum of $1.1 \mu V_{\text{rms}}$ with PGA gain equal to 1 and the minimum of $0.11 \mu V_{\text{rms}}$ equal to 24. For the sampling rate of 500 SPS, the noise is measured at the maximum of $1.51 \mu V_{\text{rms}}$ with PGA gain equal to 1 and the minimum of $0.18 \mu V_{\text{rms}}$ equal to 24. For the sampling rate of 1 kSPS, the noise is measured at the maximum of $1.92 \mu V_{\text{rms}}$ with PGA gain equal to 1 and the minimum of $0.24 \mu V_{\text{rms}}$ equal to 24.

In this study, we focused on analyzing the EEG signal features within the frequency range of 4–30 Hz, specifically targeting the theta, alpha, and beta waves. Therefore, the smallest sampling rate of 250 SPS, which guarantees a –3-dB bandwidth of 65 Hz, is enough to get the information in the EEG band we need and also help achieve the lowest noise of $0.11 \mu V_{\text{rms}}$ while using with the PGA gain of 24. The CMRR is measured at about 110 dB. These measured results adhere to the guidelines set by the IFCN for recording EEG signals in clinical settings [45].

B. Evaluation of EEG Signal of the Proposed Device

To verify the quality of the EEG signal gathered from our headband device, we established a test aiming to indicate the presence of a well-known and easily detectable neural process, the “Berger effect” (posterior alpha frequency power increases when the eyes are closed, and no more visual sensory input is processed in the occipital brain regions [26], [27], [28]). The Berger effect has already been confirmed to be well-visible with BTE EEG measuring locations [22], [26], [27]. In this test, the participant was required to do two small 1-min tasks in the resting state: eyes-open (EO) and eyes-close (EC). The test result, which is extracted using the Welch method on the collected data, revealed that the alpha-band power increased during EC, as shown in Fig. 7.

C. BTE EEG Feature Analysis

Fig. 8 shows the box-whisker plots of the distribution of EEG feature data at each measurement location for all

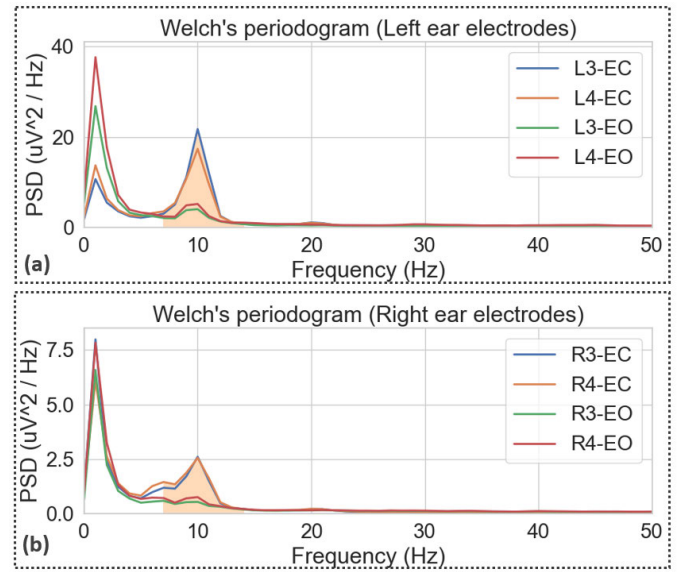


Fig. 7. PSDs of EEG signal during EC and EO resting state from the proposed BTE locations of a subject: (a) left and (b) right.

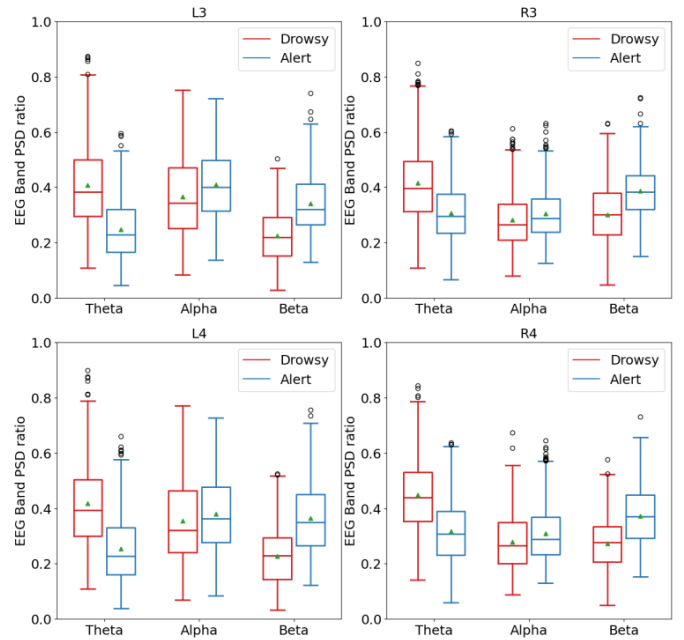


Fig. 8. Box-whisker plots of the rPSD ratio feature dataset for two states at four proposed channels: R3 (top left), R4 (top right), L3 (bottom left), and L4 (bottom right). The boxes have three lines to present the values for first quartile (bottom), median, and third quartile (top) for column data. The green triangle symbol indicates the mean value of the column data.

12 subjects. For all the channels examined, a clear trend was observed when subjects became drowsy, as the median and average values of the rPSD ratio tended to decrease in the alpha and beta bands and increase in the theta bands. In particular, the median lines of the boxes for one state (normal or drowsy) were mostly located outside or significantly separate from the boxes of the other state in the rPSD of the beta and theta bands. This indicates a high degree of differentiation in distributions between the two states in these frequency bands. While the mean and median of the rPSD ratio in the alpha

TABLE III
t-TEST TO DETERMINE WHETHER THERE IS A SIGNIFICANT DIFFERENCE IN rPSD RATIO FEATURE OF EACH EEG BAND AND CHANNELS BETWEEN THE TWO DRIVING STATES OVER ALL 12 SUBJECTS

Electrode	t-test (p-value)		
	Theta	Alpha	Beta
R3	1.62e-112	1.93e-117	9.8e-096
R4	9.57e-96	1.89e-29	1.54e-127
L3	5.3e-132	5.2e-36	2.3e-113
L4	8.26e-206	2.55e-227	1.24e-94

band during the alert state were only slightly higher than during the drowsy state, this can be attributed to the fact that alpha waves tend to decrease when a subject becomes drowsy; however, there are some moments when the subjects become significantly drowsy and they may close their eyes, causing an increase in energy of the alpha waves.

To clarify the potential difference between the drowsy and normal states in the feature dataset, the *t*-test [40] was applied on the entire feature dataset of 12 subjects. This is a widely used method for testing statistical hypotheses, where the significant statistical value (*p*-value) is used to check the distribution variation based on the test statistic value, which

$$t = \frac{(\bar{x} - \mu)}{\left(\frac{s}{\sqrt{n}}\right)} \quad (4)$$

where $\bar{x}$ is the sample mean, μ is the hypothesized mean, s is the sample standard deviation, and n is the sample size.

Table III summarizes the *p*-values of the data features of four channels for the two states. When the normal condition was less than 0.05, the *p*-values of each feature and EEG channel showed significant differences in data distribution. The *t*-test results indicate that the extracted features could be used to discriminate between the drowsy and normal states.

The high dimensionality of the data features [four channels $\times$ three rPSD ratios (theta, alpha, and beta)] made the dataset suffers from the “curse of dimensionality” [36] and prevents them from being visualized. To obtain more insights from the feature data distribution, we applied the t-distributed stochastic neighbor embedding (t-SNE) method to project the data of each subject into low-dimensional space(2-D) for visualization [36]. Fig. 9 visualizes the feature data of subject 7 using the linear SVM classifier to discriminate between the drowsy and normal states. The separation in the data distribution of the two states is considerably pronounced, which proves the feasibility of their classification in the original space.

D. Classification in MB Stage

In the MB stage, we trained the classification models and evaluated their performance on the extracted feature dataset. To guarantee the generality of the models for real-life applications, two test cases were considered: subject-dependent (SD)

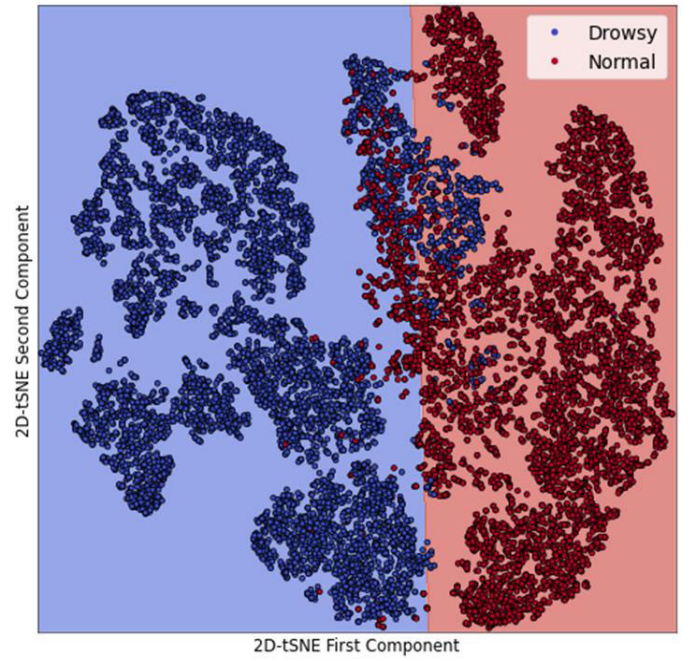


Fig. 9. Data visualization using t-SNE and SVM on the data of subject 7.

and subject-independent (SI). For the SD case, we trained models on the dataset of each subject according to the fivefold cross-validation strategy.

For the SD case, we trained models on the dataset of each subject according to the fivefold cross-validation strategy. The dataset of each subject was shuffled and split into five subsets, where one subset was used as the test set and the other four subsets were used for model training and validating (80% for training and 20% for validating). This process was iterated for each subset. The final result of each model was the average performance of the subsets. For the SI case, the leave-one-subject-out cross-validation (LOSOVCV) strategy was used. The data of one subject were taken as the test set, and the data of the rest of the subjects were used for model training and validating (80% for training and 20% for validating). We repeated this process for all subjects' data, and the average result was calculated to evaluate the performance. All training was conducted by using the optimizer Adam at a batch size of 32, a validation split rate of 0.2, and a learning rate of 0.0001.

Table IV compares the classification accuracies of SVM, MLP, and CNN for the SD and SI cases. In both cases, MLP and CNN outperformed SVM with a higher average accuracy (SD: 95.98% for MLP, 96.87% for CNN, and 91.04% for SVM; SI: 87.72% for MLP, 91.31% for CNN, and 82.43% for SVM SI). In the SD case, CNN achieved a maximum accuracy of 99.65% for subject 7. On average, the classification performances of SVM, MLP, and CNN decreased significantly when the SD case was changed to the SI case with decreases of 8.6%, 8.26%, and 5.56%, respectively. This indicates that subject variability had a remarkable effect on the DDD performance. Although CNN did not have significantly higher average accuracy than MLP in the SD case (only 0.98%), it showed considerable superiority in the SI case

TABLE IV
CLASSIFICATION ACCURACY (%) OF SVM, MLP, AND CNN IN SD AND SI CASES

Subject	Classification Accuracy (%)					
	Subject-Dependent			Subject-Independent		
	SVM	MLP	CNN	SVM	MLP	CNN
1	92.12	95.44	98.18	80.46	85.63	93.9
2	91.25	96.14	95.27	83.38	88.62	89.34
3	89.89	98.21	97.72	79.95	86.23	90.45
4	93.83	96.4	97.90	84.23	92.15	91.77
5	84.15	91.82	92.91	78.37	82.48	85.28
6	90.85	95.74	95.24	90.32	87.72	91.39
7	94.04	97.89	99.65	85.34	90.85	95.15
8	92.74	94.01	96.46	84.56	91.42	89.38
9	87.07	95.29	94.83	86.45	90.39	92.76
10	94.29	97.42	98.36	78.98	86.14	92.41
11	88.82	95.52	97.82	81.38	89.62	94.34
12	93.41	97.89	98.14	75.85	81.44	89.56
Mean	91.04	95.98	96.87	82.44	87.72	91.31
Standard Deviation	3.11	1.82	1.93	4.02	3.44	2.72

TABLE V
SUMMARY ON EVALUATING RESULT OF MLP AND CNN (%)

	Accuracy	Sensitivity	Specificity	F1-Score
MLP	94.87	96.24	93.51	94.97
CNN	96.24	97.18	95.28	96.26

(3.59%). This means that CNN was better at dealing with subject variability.

Finally, we do the evaluation on the combined dataset case. 20% of each subject dataset was split and combined to make a test set. Then, a training set was created by gathering 80% of the remaining dataset of each subject and the remaining 20% were combined to make the validation set. These three combined subsets were used to train and evaluate models to deploy on the headband. By splitting the dataset as mentioned above, the subject balance in training, validating, and testing phases can be guaranteed. We trained MLP and CNN several times until the highest accuracies of 94.87% and 96.24%, respectively, were obtained, as summarized in Table V and shown in Fig. 10. Table V reveals the superior performance of the CNN model compared to the MLP model across various performance metrics, including accuracy, sensitivity, specificity, and F1-score. Specifically, the CNN model attained an accuracy of 96.24%, while the MLP model achieved 94.87%, indicating a higher number of correct predictions overall. Moreover, the CNN model demonstrated enhanced sensitivity by correctly identifying positive instances at a rate of 97.18%, outperforming the MLP model's sensitivity of 96.24%. In addition, the CNN model exhibited superior specificity (95.28% versus 93.51%), accurately identifying negative instances. Notably, the CNN model achieved a higher F1-score of 96.26% compared to the MLP model's F1-score of 94.97%, reflecting a superior balance between precision and recall.

The CNN model yields more precise and reliable predictions, surpassing the performance of the MLP model. Then, both the trained MLP and CNN models were deployed on a headband and validated for on-device performance evaluation.

E. On-Device Testing Stage

After the previous stage, we have the well-trained original models (MLP and CNN), which are stored in the format “.h5.” The original models in “.h5” format are optimized for accuracy and performance during training and inference on high-end hardware, such as servers or GPUs. However, this format is not optimized for deployment on MCU, which has limited memory and processing power. Therefore, as mentioned above, these original models would be optimized in two ways: with and without quantization (quantized or non-quantized). In the case of nonquantized models, only the FB technique is employed to serialize the original model into a more compact and memory-friendly format using the “.h5” format. Since the weights are still stored as 32-bit floating-point values, there is no difference in values between the nonquantized model and the original model. As a result, the nonquantized model can maintain the same level of accuracy as the original model while being suitable in format and size for deployment on MCU devices. On the other hand, in the case of quantized models, the quantization method described in Section III-D is applied to convert the model's weight values from 32-bit floating-point format to 8-bit integers before utilizing the FB technique. This allows the quantized model to have a significantly smaller size and faster execution compared to the original model, albeit with a slight tradeoff in model accuracy. Then, they were all converted to C/C++ code and were loaded to the device hardware. The memory costs of these models were extracted by using a compiler (provided by Nordic Semiconductor) for analysis.

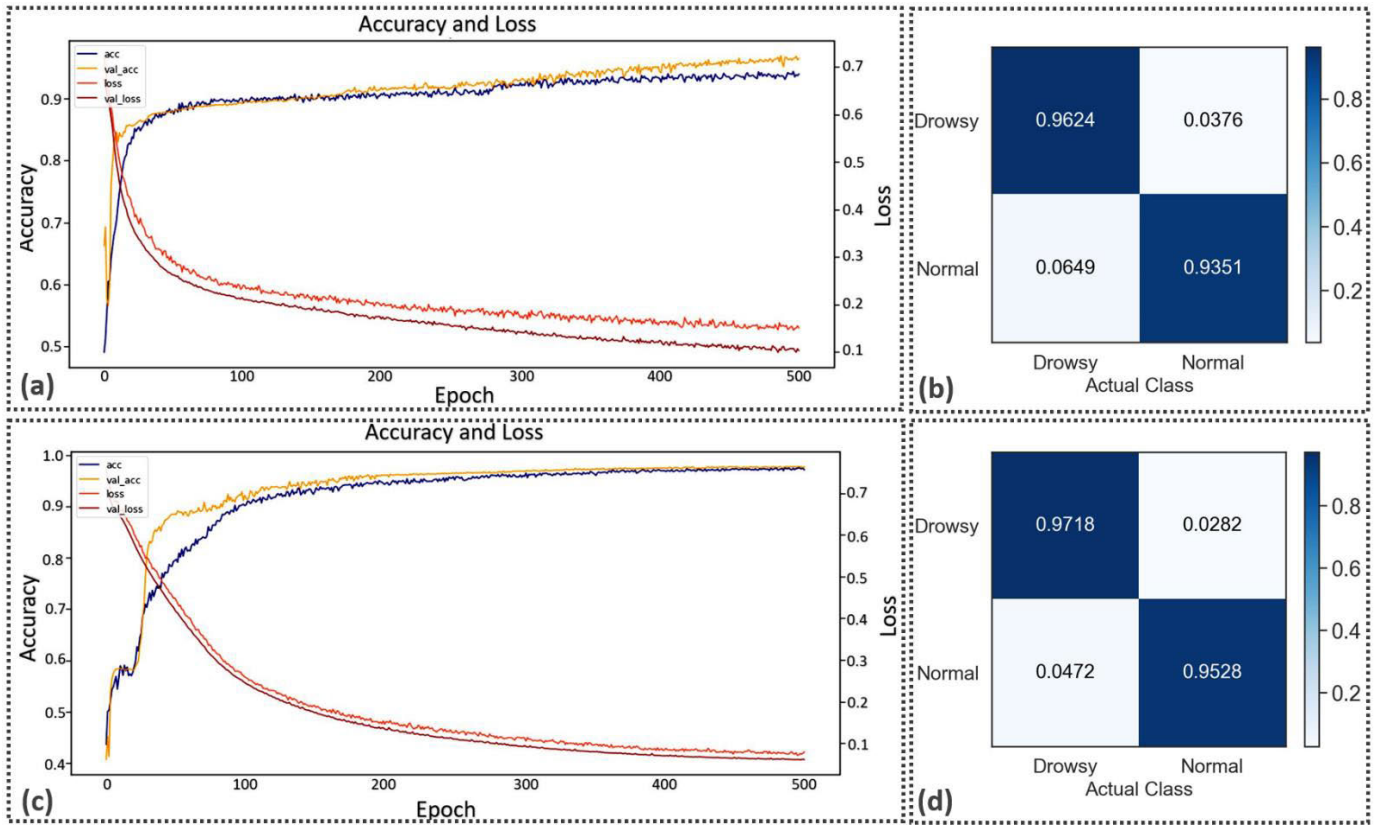


Fig. 10. Training results of (a) and (b) MLP and (c) and (d) CNN models on the dataset of all 12 subjects with accuracy–loss paragraph (blue—training accuracy, yellow—validation accuracy, orange—training loss, and red—validation loss) and confusion matrix.

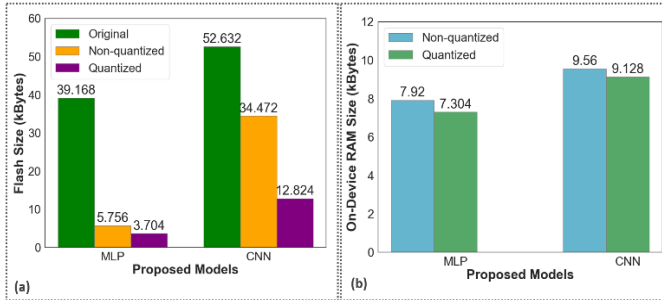


Fig. 11. (a) Flash size (kB) of two model MLP and CNN before and after optimized with and without quantization. (b) On-device RAM size (kB) of deployed models with and without quantization. The results are extracted from compiler information.

Fig. 11(a) compares the models' flash size (kB) before and after conversion for on-device deployment. Fig. 11(b) compares the on-device RAM (kB) costs of the deployed models. Overall, CNN had a larger flash size and on-device RAM size than MLP before and after optimization because of the additional convolutional and max-pooling layers in its architecture. For both models, the flash size decreased dramatically after optimization for on-device deployment and became more suitable for storage on the embedded memory-constrained device. Optimizing the model from 32-bit floating point to 8-bit integer also reduced the flash size (2.052 kB for MLP and 21.648 kB for CNN) and RAM size (0.616 kB for MLP and 0.432 kB for CNN).

TABLE VI
ACCURACY COMPARISON OF MODELS BEFORE AND AFTER CONVERTED FOR ON-DEVICE DEPLOYMENT

	Original Model (Reference)	Non-Quantized Converted Model	Quantized Converted Model
CNN	96.24%	96.24% (Degradation: 0%)	96.11% (Degradation: 0.13%)
MLP	94.87%	94.87% (Degradation: 0%)	94.6% (Degradation: 0.27%)

Then, the deployed models were run on the same test set from training to examine the degradation in accuracy before and after conversion, as presented in Table VI. Compared with the original models, both nonquantized models showed no decrease in accuracy. In contrast, the quantized models showed a negligible decrease: 0.13% for CNN and 0.27% for MLP.

The power consumption and inference latency are critical considerations for deep-learning models, especially when run in battery-constrained embedded applications [41]. Using the quantized model can reduce the power consumption and inference latency. We compared on-device computational latency and energy consumption using the Power Profiler Kit 2 (PPK2) from Nordic Semiconductor. The PPK2 connected to the Power Profiler app (PPA), monitored, and analyzed the measured parameters. For on-device DD, sensor data were collected, and two consecutive processing tasks were

TABLE VII
SUMMARY ON DURATION AND ENERGY CONSUMPTION PER ON-CHIP PROCESSING TASK*

On-chip Processing Task	Duration		Energy (μJ)
	in time (ms)	in MCU Cycles (kCycles)	
Band-pass filter	19.93	1275.52	230.8
Feature Extraction	387.47	24798.08	4219.5
On-chip Inference	Non-Quantized MLP	0.739	8.646
	Quantized MLP	0.258	2.607
	Non-Quantized CNN	28.66	351.219
	Quantized CNN	25.44	277.8

* Note that these results are calculated considering maximum operating frequency of 64MHz and 3.3V

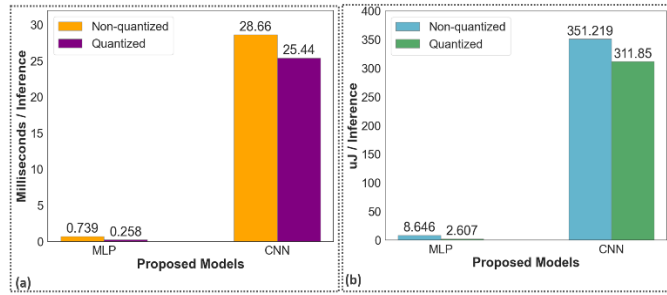


Fig. 12. On-device comparison between two models MLP and CNN with and without quantization, including two parameters: (a) average inference latency (ms/inference) and (b) average energy consumption per inference ($\mu\text{J}/\text{inference}$) after doing the measurement 50 times (64 MHz@3.3 V).

performed on an 8.192-s four-channel input EEG data epoch: SP&FE (bandpass filter and FE) and classification. MLP and CNN models were compared, and performance information was extracted using the PPK2 and PPA.

Table VII indicates the summary on duration and energy consumption per on-chip processing task. The bandpass filter task took 19.93 ms (1275.2 kCycles) with an energy cost of 230 μJ . Meanwhile, the most resource-intensive on-device processing task was features extraction, which had a computational latency of 387.47 ms (24 798.08 kCycles) and consumed 4219.5 μJ . Fig. 12 compares the on-device performances of MLP and CNN according to two parameters: the average inference latency (ms/inference) and average energy consumption per inference ($\mu\text{J}/\text{inference}$) after 50 measurements. Note that the measurements were conducted at room temperature (22 °C) and the MCU frequency is at 64 MHz. Fig. 12(a) indicates that CNN took much longer than MLP to run inference with both optimization techniques. For both MLP and CNN, the quantization technique reduced the computational cost by 0.481 and 3.22 ms/inference, respectively. Fig. 12(b) shows that CNN had a higher energy consumption than MLP at 342.573 μJ higher without quantization and 309.243 μJ higher with quantization. Using quantization reduced the power consumption for both models: 3.3 times lower for MLP and 1.12 times lower for CNN.

The experimental results presented above clearly demonstrate that the proposed CNN model has slightly higher power consumption, computational speed, and memory cost compared to MLP, regardless of whether a quantization technique is used. Although the differences in these parameters are minimal, they are outweighed by the significantly improved

TABLE VIII
SUMMARY ON POWER CONSUMPTION OF THE PROPOSED HEADBAND (mW)

Operating Mode	Digital	Radio	Analog	Total
REC Mode (CI = 7.5ms)	4 channels	13.134	3.56	44.363
	8 channels	14.348	3.99	67.821
DDD Mode (CI = 7.5ms) *	12.144	1.617	44.363	58.124
DDD Mode (CI = 100ms) *	12.144	0.924	44.363	57.431

*Run consecutively without sleep (4 channels)

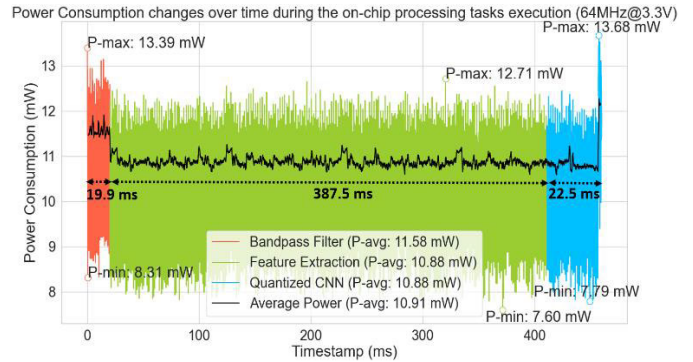


Fig. 13. Power consumption changes over time during the on-chip processing task execution (64 MHz@3.3 V).

accuracy achieved by the CNN model in detecting drowsiness, especially when dealing with subject variability. Thus, despite the slightly higher resource requirements, the CNN model is a more effective and reliable choice for DD. Therefore, together with considering the significant benefits of using the quantization technique at the expense of a negligible reduction in accuracy (only 0.13%), we have selected a quantized CNN model for on-device DDD applications. Fig. 13 shows the changes over time in power consumption during the execution of the algorithm on the proposed device.

Finally, we combined our headband device with a battery of 300 mAh and measured the total power consumption of the device when working in two modes: recording mode and DD mode. Table VIII provides an overview of the board's power consumption in different operational modes. During the testing phase in recording mode, the device was operating with BLE CI of 7.5 ms as previously mentioned, considering two

TABLE IX
COMPARISON WITH PREVIOUS WORKS ON EEG-BASED DDD SYSTEM

Works	No. of subjects	Features	Methodology	Results	EEG Device	Process Platform	Battery Duration & Power
[46]	14	+ 7chn scalp-EEG (Raw Signal)	+ Using a CNN model for classification with Data Augmentation	2-states Mixed dataset with avg. accuracy: 90.14%	Emotive EPOC+ headset	On PC	640mAh-3.7V: +Stream Data: 6h/14chn (395mW)
[47]	12	+ 19chn scalp-EEG signal (ApEn, SamEn, Modified SamEn)	+ Using a SVM for classifier with RELIEFF for feature selection	2-states 10-fold evaluation with avg. accuracy: 86.84%	Full-scalp-cover headset	On PC	---
[53]	12	+ 2chn Occipital-EEG (α wavelet energy) +VEOG (wavelet energy)	+Using CWT with Haar for wavelet energy extraction. +CWGAN for data augmentation +LSTM-based model as classifier	2 states with avg. accuracy: +LOSOV: 98.14%	ESI NeuroScan	On PC	---
[55]	14	+ 2chn scalp-EEG: Fz and Pz (Raw Signal)	+ Using Capsule Neural Network with Data Augmentation	2 states with avg. accuracy: + 5-fold cross-validation: 91.24%	---	On PC	---
[51]	18	+ 4chn Temporal-EEG (1st and 2nd AR coefficients and LRSSV)	+ Using SVM classifier applied on 3 selected features dataset.	2 states with avg. accuracy: +LOSOV: 78.3%	Muse-2	On PC	290mAh-3.7V: +Stream Data: 5h/5chn (214mW)
[49]	10	+ 2chn Occipital-EEG (Blink Dur., α Power) + IMU (Gesture)	+ Using threshold values for each feature, then combine to detect levels of drowsiness.	5 states with avg. accuracy: 95.2%	Custom Device: +MCU: STM32F407 +AFE: ADS1298 +IMU: MPU9150 +Standard Bluetooth	On MCU On PULP	200mAh-3.7V: +Process on MCU: 6h (123mW) +Process on PULP: 46h (16mW)
[50]	3	+ 3chn Occipital-EEG (Blink Dur., CAP, CAW) + IMU (Gesture)	+ Using PCA to merge 3-D EEG signals into 1-D vector and then apply feature extraction. + Using k-means-based algorithm for classifier	5 states with avg. accuracy: + Offline: 83% + Online: 83%	Custom Device: +SoC: nRF52832 (incl. MCU & BLE) +PULP: Mr.Wolf +AFE: ADS1298 +IMU: MPU9150	On PULP	200mAh-3.7V: +Compute on PULP: 60h (12.33mW)
[52]	19	+ 8chn BTE EEG/EOG/EMG and EDA (24 Temporal features, 28 Spectral features and 4 Non-linear features)	+ Use three feature selection methods (Recursive Feature Elimination (RFE), L1-based, and Tree-based) to choose the best features. + Using a hierarchical stack of three base classifiers (1st – Random Forest, 2nd – Adaboost, 3rd – SVM with RBF)	2-states detection (Microsleep or not) with avg. Precision, Recall, F1-score: +LOSOV: 75%-58%-81% +10-fold Cross Validation: 90%-94%-90%	Custom Device: +MCU: MSP430 +AFE: ADS1299 +SD Card + Standard Bluetooth	On PC	600mAh-3.7V: + Stream Data: 9.2h/8chn (241.3mW)
This study	12	+ 4chn BTE EEG (Relative power spectral density (PSD) of θ , α , and β band)	+ CNN model for classification on feature extracted dataset	2 states with avg. accuracy: +LOSOV: 91.31% +Mixed dataset split with subject balance: 96.24%	Custom Device: + SoC: nRF52840 (incl. MCU & BLE) + AFE: ADS1299	On MCU	300mAh-3.7V: + Stream Data (REC Mode): 12.9h/8chn (86.159mW), 18.2h/4chn (61.057mW) + Compute on MCU (on-chip DDD Mode): 19.33h (58.124mW)

options: four channels and eight channels, resulting in power consumption levels of approximately 61.057 mW (equivalent to 18.2 h of battery duration) and 86.159 mW (equivalent to 12.9 h of battery duration), respectively. It is revealed

that ADS1299 is the component that consumes the most power in both options, with power consumption levels of 44.363 mW for four channels and 67.821 mW for eight channels.

In the DD mode, using a quantized CNN model, our device employed four channels and underwent testing with two BLE CI options: 7.5 and 100 ms. The corresponding power consumption readings were 58.124 mW (19.1 h of battery duration) and 57.431 mW (19.33 h of battery duration). It is noted that the device runs the process for DD continuously without sleep phase on this test. It revealed that increasing the BLE CI from 7.5 to 100 ms in the DD mode led to a reduction in power consumption by approximately 1.307 mW. The packet counter order was also examined to ensure that there is not any problem related to packet loss in both options. The conventional EEG-based DDD system mentioned earlier typically relies on computer-based hardware for carrying out computational stages. Here, the wearable EEG devices of these systems function similar to our device in recording mode, solely capturing raw EEG data and transmitting it to the PC-based hardware. Therefore, the outcomes also indicated that by implementing on-chip DD, which reduces the number of bytes in the BLE packet as well as the transmission frequency, the device can reduce energy consumption by up to 6%, which helps extend the battery duration from 18.18 to 19.33 h.

F. Compare to Previous Works

Numerous previous studies have investigated the use of EEG signals for DD. Most of these studies focus on detecting drowsiness based on EEG using many electrodes on the scalp. This makes EEG devices becoming large, cumbersome, and insufficiently unobtrusive for use in real-life applications, as shown in [46] and [47]. Several efforts have been made to improve this by reducing the number of measurement points on the head, making wearable EEG devices more compact, as demonstrated in articles [49], [50], [51], [53], and [55]. To improve the detection performance, Kartsch et al. [49] and [50] utilized additional Inertial Measurement Unit (IMU) data, while Jiao et al. [53] incorporated additional vertical electrooculogram (VEOG) data. The occipital region was widely used in these studies because of its association with the alpha rhythm (8–13 Hz), which is one of the prominent indicators of drowsiness. However, as mentioned in Section I, when applied in the situation of driving vehicles, the electrode placed in the occipital region can make the driver uncomfortable while relying on the seat. BTE EEG can be considered as a novel solution, which is being explored by a few recent research studies, including our own one. One such study by Pham et al. [52] introduced using a method using a hierarchical stack of three base ML classifiers to detect microsleep event, which happened in an extremely drowsy state. However, this study system utilized computer-based hardware to perform computations for detection algorithm based on the data received from the wearable device, resulting in high energy consumption and reducing real-time detection capability. In addition, this also makes the system bulky and less user-friendly. This approach is also employed in almost other conventional DD research, such as [46] and [51]. Developing wearable devices with integrated on-device detection capabilities can address these limitations. Some studies have explored this solution, such as [49] and [50]. Kartsch et al. [49] and [50] improved

computing capabilities and reduced energy consumption costs impressively by using a parallel ultralow-power (PULP) processor for detection algorithm instead of the MCU in the wearable device. However, it is important to note that this approach may introduce design complexity and larger size of hardware and increase the device's cost, and in addition, PULP processors are also not yet widely available in the market for easy procurement. However, none of them utilized deep-learning approaches. Deep-learning approaches were implemented for EEG-based DD research in [46], [53], and [55]. However, the deep-learning-based solutions proposed in these studies are also still implemented on computer-based hardware instead of on EEG devices. Our study proposes a solution to address these challenges with a novel low-power wearable headband. This headband is designed for on-device DDD with BTE EEG, offering a lightweight and affordable option. It includes signal processing features and a compact deep-learning model while ensuring comfort for long-term use and facilitating wireless data collection. Table IX presents a detailed comparison of the relevant information between our proposed system and previous studies on the DDD system.

V. CONCLUSION

We developed an on-device DDD MCU-based system using a BTE EEG signal. The wearable headband device was designed to be convenient for users in daily life and to provide good BTE EEG signals. The analysis results showed that the rPSD ratio of three EEG bands (theta, alpha, and beta) at four measurement locations (8.192-s epoch) have high potential for DDD applications. Two deep-learning models (CNN and MLP) were implemented to use these extracted features for classification, and their performances were compared with that of SVM. CNN and MLP outperformed the SVM model in both SD and SI cases. Although the CNN was only slightly more accurate than MLP in the SD case, it was significantly more accurate in the SI case, which indicates that it can better handle subject variability. When the two models were trained on a dataset of all subjects, CNN again performed better than MLP.

TFLM was used with two optimization techniques (with and without quantization) to convert the trained CNN and MLP into C/C++ code to run on the embedded device for on-device tests. For both MLP and CNN, quantization resulted in better optimization in terms of the flash size (kB), RAM size (kB), average inference time (ms), and power consumption (μ C) with a negligible decrease in accuracy. With both optimization methods, MLP showed a dramatic drop in on-device performance parameters compared with CNN. These results demonstrate the feasibility of using the BTE EEG signal and on-chip NN with an MCU-based device for DDD applications. We believe that these findings will inspire more real-world on-device DDD applications based on BTE EEG.

Although the performance of the proposed system was promising, further studies are still needed. For example, using smaller BTE EEG epochs can make the system act in more real time. The ensemble learning method, which gives predictions using multiple models, may be a better candidate to enhance the DDD performance. Increasing the number of detected

drowsiness levels can make the application more detailed and practical. Our device currently experiences significant processing time during FE, which is the most time-consuming task. However, we aim to improve this aspect in our future work. Our focus will be on developing a deep-learning model capable of handling the raw signal, eliminating the need for manual FE. By doing so, we anticipate making the system more real-time, enhancing its overall performance. It is noted that we conducted our research with a limited number of participants (12 subjects), which may impact the generalizability of our findings. By acknowledging this need for future research and validation, we aim to strengthen the credibility of our proposed approach. Furthermore, within this study, the observers adhered to Wierwille's scale (presented in Table I) to assess the levels of alertness/drowsiness. While it was inevitable that instances could arise where their judgments might not be entirely precise, we attempted to mitigate this concern by implementing the proposed voting mechanism, which was outlined with meticulous step-by-step instructions. Moving forward, we intend to enhance our proposed methodology's reliability by incorporating alternative approaches.

Another issue addressed in this study concerns SP. In this investigation, we employed a basic bandpass filter to eliminate unwanted signals beyond the frequency range of 1–40 Hz. Furthermore, participants were instructed to maintain a comfortable seated position and minimize unnecessary movements that could introduce noise interference in the experimental outcomes. The primary objective of utilizing this simple filter during SP was to decrease the computation time involved, thus allowing more time for other vital tasks such as FE or the execution of ML models. This choice aimed to strike a balance between achieving accuracy and accommodating the real-time nature of the system. However, it is important to acknowledge that this basic filter has limitations when processing other artifacts caused by participants during the experiment, such as eye blinks. Consequently, future research will concentrate on implementing advanced and efficient SP techniques, such as blind source separation (BSS) or independent component analysis (ICA), to reduce noise stemming from body movements, eye blinks, muscle activity, cardiac signals, and power lines.

Besides, we would like to acknowledge a limitation regarding the use of the 8.192-s epoch with overlaps of 50% to augment the dataset for training ML models. While the time difference between two adjacent epochs is 4.096 s (1024 data points), they may not exhibit significant variations on time scales. Consequently, this may result in highly correlated data, potentially enhancing the system's performance. To address the potential issue of highly correlated data in our study, we employed the cross-validation technique to train the models with two distinct test cases: SD and SI. In addition, in the architecture of our proposed NN models, MLP and CNN, we incorporated dropout layers with a dropout rate of 50% to mitigate overfitting and discourage heavy reliance on correlated features. In future work, we will avoid using the overlapping technique to prevent the occurrence of a highly correlated dataset issue. Furthermore, we will consider applying regularization techniques such as $L1$ or $L2$

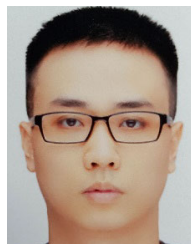
regularization on our ML models. Regularization can also aid in reducing overfitting and minimizing the model's dependence on correlated features.

In addition, as part of our future work, we plan to incorporate walk-forward cross validation (also referred to as rolling cross validation) [54] for assessing the effectiveness of our time series forecasting model. This technique can enhance the model's real-time performance by iteratively training it on a subset of the available data, generating predictions for the subsequent time period, evaluating its performance, and advancing the training set by one time step to include the latest data point. Our device currently experiences significant processing time during FE, which is the most time-consuming task. However, we aim to improve this aspect in our future work. Our focus will be on developing a deep-learning model capable of handling the raw signal, eliminating the need for manual FE. By doing so, we anticipate making the system more real time, enhancing its overall performance. We will try to address the above limitations in our future work. In future work, we also aim to undertake an in-depth investigation to produce a comprehensive academic report on the correlation between the three available sampling rates of device and their implications for DD performance.

REFERENCES

- [1] *Drowsy Driving 2015; NHTSA's National Center for Statistics and Analysis*, Nat. Highway Traffic Saf. Admin., Washington, DC, USA, 2017.
- [2] J. He, W. Choi, Y. Yang, J. Lu, X. Wu, and K. Peng, "Detection of driver drowsiness using wearable devices: A feasibility study of the proximity sensor," *Appl. Ergonom.*, vol. 65, pp. 473–480, Nov. 2017.
- [3] P. M. Forsman, B. J. Vila, R. A. Short, C. G. Mott, and H. P. A. Van Dongen, "Efficient driver drowsiness detection at moderate levels of drowsiness," *Accident Anal. Prevention*, vol. 50, pp. 341–350, Jan. 2013.
- [4] S. Abtahi, B. Hariri, and S. Shirmohammadi, "Driver drowsiness monitoring based on yawning detection," in *Proc. IEEE Int. Instrum. Meas. Technol. Conf.*, May 2011, pp. 1–4.
- [5] A. Dasgupta, D. Rahman, and A. Routray, "A smartphone-based drowsiness detection and warning system for automotive drivers," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 11, pp. 4045–4054, Nov. 2019.
- [6] R. O. Mbouna, S. G. Kong, and M.-G. Chun, "Visual analysis of eye state and head pose for driver alertness monitoring," *IEEE Trans. Intell. Transp. Syst.*, vol. 14, no. 3, pp. 1462–1469, Sep. 2013.
- [7] S. Jung, H. Shin, and W. Chung, "Driver fatigue and drowsiness monitoring system with embedded electrocardiogram sensor on steering wheel," *IET Intell. Transp. Syst.*, vol. 8, no. 1, pp. 43–50, Feb. 2014.
- [8] L. Boon-Leng, L. Dae-Seok, and L. Boon-Giin, "Mobile-based wearable-type of driver fatigue detection by GSR and EMG," in *Proc. IEEE Region 10 Conf. (TENCON)*, Nov. 2015, pp. 1–4.
- [9] R. N. Khushaba, S. Kodagoda, S. Lal, and G. Dissanayake, "Driver drowsiness classification using fuzzy wavelet-packet-based feature-extraction algorithm," *IEEE Trans. Biomed. Eng.*, vol. 58, no. 1, pp. 121–131, Jan. 2011.
- [10] G. Li, B.-L. Lee, and W.-Y. Chung, "Smartwatch-based wearable EEG system for driver drowsiness detection," *IEEE Sensors J.*, vol. 15, no. 12, pp. 7169–7180, Dec. 2015.
- [11] A. Craig, Y. Tran, N. Wijesuriya, and H. Nguyen, "Regional brain wave activity changes associated with fatigue," *Psychophysiology*, vol. 49, no. 4, pp. 574–582, Apr. 2012.
- [12] G. Borghini, L. Astolfi, G. Vecchiato, D. Mattia, and F. Babiloni, "Measuring neurophysiological signals in aircraft pilots and car drivers for the assessment of mental workload, fatigue and drowsiness," *Neurosci. Biobehavioral Rev.*, vol. 44, pp. 58–75, Jul. 2014.
- [13] J. LaRocco, M. D. Le, and D.-G. Paeng, "A systemic review of available low-cost EEG headsets used for drowsiness detection," *Frontiers Neuroinform.*, vol. 14, Oct. 2020.

- [14] C.-S. Wei, Y.-T. Wang, C.-T. Lin, and T.-P. Jung, "Toward drowsiness detection using non-hair-bearing EEG-based brain-computer interfaces," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 26, no. 2, pp. 400–406, Feb. 2018.
- [15] R. Foong, K. K. Ang, Z. Zhang, and C. Quek, "An iterative cross-subject negative-unlabeled learning algorithm for quantifying passive fatigue," *J. Neural Eng.*, vol. 16, no. 5, Aug. 2019, Art. no. 056013.
- [16] I. Stancin, M. Cifre, and A. Jovic, "A review of EEG signal features and their application in driver drowsiness detection systems," *Sensors*, vol. 21, no. 11, p. 3786, May 2021.
- [17] N. S. Karuppusamy and B.-Y. Kang, "Multimodal system to detect driver fatigue using EEG, gyroscope, and image processing," *IEEE Access*, vol. 8, pp. 129645–129667, 2020.
- [18] Y. Jiang, Y. Zhang, C. Lin, D. Wu, and C.-T. Lin, "EEG-based driver drowsiness estimation using an online multi-view and transfer TSK fuzzy system," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 3, pp. 1752–1764, Mar. 2021.
- [19] K. B. Mikkelsen, S. L. Kappel, D. P. Mandic, and P. Kidmose, "EEG recorded from the ear: Characterizing the ear-EEG method," *Frontiers Neurosci.*, vol. 9, p. 438, Nov. 2015.
- [20] P. Kidmose, D. Looney, M. Ungstrup, M. L. Rank, and D. P. Mandic, "A study of evoked potentials from ear-EEG," *IEEE Trans. Biomed. Eng.*, vol. 60, no. 10, pp. 2824–2830, Oct. 2013.
- [21] D. Looney et al., "The in-the-ear recording concept: User-centered and wearable brain monitoring," *IEEE Pulse*, vol. 3, no. 6, pp. 32–42, Nov. 2012.
- [22] S.-I. Choi et al., "On the feasibility of using an ear-EEG to develop an endogenous brain-computer interface," *Sensors*, vol. 18, no. 9, p. 2856, Aug. 2018.
- [23] G. Li and W.-Y. Chung, "Combined EEG-gyroscope-tDCS brain machine interface system for early management of driver drowsiness," *IEEE Trans. Human-Mach. Syst.*, vol. 48, no. 1, pp. 50–62, Feb. 2018.
- [24] P. Warden and D. Situnayake, *Tiny ML: Machine Learning with Tensorflow Lite on Arduino and Ultra-Low-Power Microcontrollers*. Sebastopol, CA, USA: O'Reilly Media, 2019.
- [25] Texas Instruments. (Jul. 2012). *ADS1299-x Low-Noise, 4-, 6-, 8-Channel, 24-Bit, Analog-to-Digital Converter for EEG and Biopotential Measurements*. [Online]. Available: <http://www.ti.com/lit/ds/symlink/ads1299.pdf>
- [26] M. G. Bleichner and S. Debener, "Concealed, unobtrusive ear-centered EEG acquisition: cEEGrids for transparent EEG," *Frontiers Human Neurosci.*, vol. 11, p. 163, Apr. 2017.
- [27] M. G. Bleichner and S. Debener, "Independent component decomposition of around ear EEG data to detect artifacts," in *Proc. IEEE Int. Conf. Syst., Man Cybern. (SMC)*, Oct. 2019, pp. 3631–3634.
- [28] K. Kirschfeld, "The physical basis of alpha waves in the electroencephalogram and the origin of the 'Berger effect,'" *Biol. Cybern.*, vol. 92, no. 3, pp. 177–185, Mar. 2005.
- [29] NHTSA. *Drowsy Driving: Avoid Falling Asleep Behind the Wheel*. Accessed: Jun. 24, 2022. [Online]. Available: <https://www.nhtsa.gov/risky-driving/drowsy-driving>
- [30] Wikipedia. (Aug. 23, 2022). *Circadian Rhythm*. [Online]. Available: https://en.wikipedia.org/wiki/Circadian_rhythm
- [31] A. Shahid et al., "Karolinska sleepiness scale (KSS)," in *STOP, THAT and One Hundred Other Sleep Scales*. New York, NY, USA: Springer, 2011, pp. 209–210.
- [32] K. Kaida et al., "Validation of the Karolinska sleepiness scale against performance and EEG variables," *Clin. Neurophysiol.*, vol. 117, no. 7, pp. 1574–1581, Jul. 2006.
- [33] H. Mårtensson, O. Keelan, and C. Ahlström, "Driver sleepiness classification based on physiological data and driving performance from real road driving," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 2, pp. 421–430, Feb. 2019.
- [34] W. W. Wierwille and L. A. Ellsworth, "Evaluation of driver drowsiness by trained raters," *Accident Anal. Prevention*, vol. 26, no. 5, pp. 571–581, Oct. 1994.
- [35] M. Zare, H. R. Pourghasemi, M. Vafakhah, and B. Pradhan, "Landslide susceptibility mapping at Vaz Watershed (Iran) using an artificial neural network model: A comparison between multilayer perceptron (MLP) and radial basic function (RBF) algorithms," *Arabian J. Geosci.*, vol. 6, no. 8, pp. 2873–2888, Aug. 2013.
- [36] A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems*. Sebastopol, CA, USA: O'Reilly Media, 2019.
- [37] TensorFlow. (Jul. 22, 2022). *TensorFlow Lite for Microcontrollers*. [Online]. Available: <https://www.tensorflow.org/lite/microcontrollers>
- [38] TensorFlow. (Jun. 11, 2022). *Convert TensorFlow Models*. [Online]. Available: https://www.tensorflow.org/lite/models/convert/convert_models
- [39] B. Jacob et al., "Quantization and training of neural networks for efficient integer-arithmetic-only inference," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2704–2713.
- [40] T. K. Kim, "T test as a parametric statistic," *Korean J. Anesthesiol.*, vol. 68, no. 6, pp. 540–546, 2015.
- [41] B. Moons, K. Goetschalckx, N. Van Berckelaer, and M. Verhelst, "Minimum energy quantized neural networks," in *Proc. 51st Asilomar Conf. Signals, Syst., Comput.*, Oct. 2017, pp. 1921–1925.
- [42] T. Hwang, M. Kim, S. Hong, and K. S. Park, "Driver drowsiness detection using the in-ear EEG," in *Proc. 38th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Aug. 2016, pp. 4646–4649.
- [43] T. Nakamura, Y. D. Alqurashi, M. J. Morrell, and D. P. Mandic, "Automatic detection of drowsiness using in-ear EEG," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2018, pp. 1–6.
- [44] S. Hong, H. Kwon, S. H. Choi, and K. S. Park, "Intelligent system for drowsiness recognition based on ear canal electroencephalography with photoplethysmography and electrocardiography," *Inf. Sci.*, vol. 453, pp. 302–322, Jul. 2018.
- [45] M. R. Nuwer et al., "IFCN standards for digital recording of clinical EEG," *Electroencephalogr. Clin. Neurophysiol.*, vol. 106, no. 3, pp. 259–261, Mar. 1998.
- [46] S. Chaabene, B. Bouaziz, A. Boudaya, A. Hökelmann, A. Ammar, and L. Chaari, "Convolutional neural network for drowsiness detection using EEG signals," *Sensors*, vol. 21, no. 5, p. 1734, Mar. 2021.
- [47] A. Chaudhuri and A. Routray, "Driver fatigue detection through chaotic entropy analysis of cortical sources obtained from scalp EEG signals," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 1, pp. 185–198, Jan. 2020.
- [48] M. Guermandi, A. Cossetini, S. Benatti, and L. Benini, "A wireless system for EEG acquisition and processing in an earbud form factor with 600 hours battery lifetime," in *Proc. 44th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2022, pp. 3139–3145.
- [49] V. J. Kartsch, S. Benatti, P. D. Schiavone, D. Rossi, and L. Benini, "A sensor fusion approach for drowsiness detection in wearable ultra-low-power systems," *Inf. Fusion*, vol. 43, pp. 66–76, Sep. 2018.
- [50] V. Kartsch, S. Benatti, M. Guermandi, F. Montagna, and L. Benini, "Ultra low-power drowsiness detection system with BioWolf," in *Proc. 9th Int. IEEE/EMBS Conf. Neural Eng. (NER)*, San Francisco, CA, USA, Mar. 2019, pp. 1187–1190, doi: [10.1109/NER.2019.8717070](https://doi.org/10.1109/NER.2019.8717070).
- [51] S. Gangadharan K and A. P. Vinod, "Drowsiness detection using portable wireless EEG," *Comput. Methods Programs Biomed.*, vol. 214, Feb. 2022, Art. no. 106535.
- [52] N. Pham et al., "Detection of microsleep events with a behind-the-ear wearable system," *IEEE Trans. Mobile Comput.*, vol. 22, no. 2, pp. 841–857, Feb. 2023, doi: [10.1109/TMC.2021.3090829](https://doi.org/10.1109/TMC.2021.3090829).
- [53] Y. Jiao, Y. Deng, Y. Luo, and B.-L. Lu, "Driver sleepiness detection from EEG and EOG signals using GAN and LSTM networks," *Neurocomputing*, vol. 408, pp. 100–111, Sep. 2020.
- [54] R. M. Stein, "Benchmarking default prediction models: Pitfalls and remedies in model validation," Moody's KMV, New York, NY, USA, Tech. Rep. 20305, 2002.
- [55] L. Guarda, J. E. Tapia, E. L. Drogue, and M. Ramos, "A novel capsule neural network based model for drowsiness detection using electroencephalography signals," *Expert Syst. Appl.*, vol. 201, Sep. 2022, Art. no. 116977.



Ha-Trung Nguyen received the B.E. degree in electronics and telecommunications from the Hanoi University of Science and Technology, Hanoi, Vietnam, in 2020. He is currently pursuing the master's degree with the Department of Artificial Intelligence Convergence, Pukyong National University, Busan, South Korea.

His research interests include wearable healthcare devices, electroencephalogram (EEG)-based brain-computer interface systems, computer vision, and machine learning.



Ngoc-Dau Mai received the B.E. degree in electronics and telecommunications from the Hanoi University of Science and Technology, Hanoi, Vietnam, in 2021. He is currently pursuing the master's degree with the Department of Artificial Intelligence Convergence, Pukyong National University, Busan, South Korea.

His research interests include the development of wearable healthcare devices, signal processing, electroencephalogram (EEG)-based affective computing systems, and machine learning.



Boon Giin Lee (Senior Member, IEEE) received the B.I.T. degree from Multimedia University, Melaka, Malaysia, in 2007, the M.S. degree from Dongseo University, Busan, South Korea, in 2009, and the Ph.D. degree from Pukyong National University, Busan, in 2013.

He worked with Changsung Ace Company Ltd., Seoul, South Korea, as a Software Engineer for a year. After this, he served as an Assistant Professor with Keimyung University, Daegu, South Korea, for six years. He joined

the University of Nottingham Ningbo China, Ningbo, China, in 2019. Dr. Lee research interests covered a number of areas, including human-computer interaction, sensors application, and computer vision.



Wan-Young Chung (Senior Member, IEEE) received the B.S. and M.S. degrees in electronic engineering from Kyungpook National University, Daegu, South Korea, in 1987 and 1989, respectively, and the Ph.D. degree in sensor engineering from Kyushu University, Fukuoka, Japan, in 1998.

He was an Assistant Professor with Semyung University, Jecheon, South Korea, from 1993 to 1999. He was an Associate Professor with Dongseo University, Busan,

South Korea, from 1999 to 2008. He has been a Full Professor with the Department of Electronic Engineering, Pukyong National University, Busan, since 2008. His areas of interest include ubiquitous healthcare, wireless sensor network applications, and gas sensors.